Below is a complete **Week 5 – Visualization Interview Question Bank** covering the topics shown in your image. I have separated it into:

1. **Definition / Concept-based Interview Questions with Answers**
2. **Scenario-based Interview Questions with Answers**

The answers are written so they can be spoken directly in an interview.

PART 1 — DEFINITION-BASED INTERVIEW QUESTIONS & ANSWERS

A. Power BI Introduction

1. What is Power BI?

Answer:
Power BI is a Business Intelligence and data visualization platform from Microsoft. It allows us to connect to different data sources, transform and model data, create interactive reports and dashboards, and share insights with users.

---

2. What are the major components of Power BI?

Answer:

The main components are:

- **Power BI Desktop** – Used to develop reports and data models.
- **Power BI Service** – Cloud platform used for publishing, sharing, refreshing, dashboards and alerts.
- **Power BI Mobile** – Used to view reports on mobile devices.
- **Power BI Gateway** – Connects Power BI Service to on-premises data sources.
- **Power BI Report Server** – Used for on-premises report hosting.

---

3. What is Power BI Desktop?

Answer:
Power BI Desktop is the development application used to connect to data, transform the data, create relationships, write DAX calculations and build visual reports.

---

4. What is Power BI Service?

Answer:
Power BI Service is the cloud-based version of Power BI used to publish reports, create dashboards, schedule refreshes, configure alerts and share content with users.

---

5. What is the difference between Power BI Desktop and Power BI Service?

Answer:

Power BI Desktop is mainly used for:

- Data transformation
- Data modeling
- DAX
- Report development

Power BI Service is mainly used for:

- Publishing
- Sharing
- Dashboard creation
- Scheduled refresh
- Alerts
- Collaboration

B. Connecting to a Data Source

6. What types of data sources can Power BI connect to?

Answer:
Power BI can connect to many data sources including:

- Excel
- CSV
- SQL Server
- MySQL
- PostgreSQL
- Oracle
- Snowflake
- Google BigQuery
- Azure SQL
- SharePoint
- Web APIs
- JSON files

- Cloud platforms

---

7. What is a data connector in Power BI?

Answer:
A data connector is an interface provided by Power BI that allows Power BI to communicate with an external data source such as SQL Server, Snowflake or BigQuery.

---

8. What are Import mode and DirectQuery mode?

Answer:

Import mode copies data into the Power BI data model.

DirectQuery leaves the data in the original database and sends queries to the database whenever users interact with the report.

---

9. What is the difference between Import and DirectQuery?

Import	DirectQuery
Data is loaded into Power BI	Data remains in source
Usually faster reports	Depends on source performance
Requires refresh	Can query current source data
Supports more modeling features	Some limitations may exist

---

10. What is a Live Connection?

Answer:
A Live Connection allows Power BI to connect directly to an existing semantic model, such as Analysis Services or a published Power BI semantic model, without importing the model into the report.

C. Importing Data

11. What happens when data is imported into Power BI?

Answer:
Power BI reads data from the source, optionally transforms it using Power Query, compresses it using the VertiPaq engine and stores it inside the Power BI semantic model.

---

12. What is the VertiPaq engine?

Answer:
VertiPaq is Power BI's in-memory columnar storage engine. It compresses data efficiently and provides fast analytical query performance.

---

13. What is the difference between Load and Transform Data?

Answer:

Load directly loads selected data into Power BI.

Transform Data opens Power Query Editor where we can clean and modify the data before loading it.

D. Data Manipulation

14. What is data manipulation?

Answer:
Data manipulation means changing or restructuring data so that it becomes suitable for analysis. Examples include filtering rows, creating columns, replacing values and combining tables.

---

15. What are common data manipulation operations?

Answer:

Common operations include:

- Filtering
- Sorting
- Grouping
- Renaming columns
- Replacing values
- Splitting columns
- Merging tables
- Appending tables
- Creating calculated columns

E. Data Transformation

16. What is data transformation?

Answer:
Data transformation is the process of converting raw data into a clean and useful structure before analysis.

For example:

Raw Date

21/08/2026

→

Year

2026

Month

August

Day

21

---

17. What tool is used for data transformation in Power BI?

Answer:
Power Query Editor is mainly used for data transformation.

---

18. What language does Power Query use?

Answer:
Power Query uses the **M language**, also called Power Query M.

---

19. What is Power Query?

Answer:
Power Query is Power BI's data preparation and transformation engine used to connect, clean, combine and reshape data before loading it into the data model.

F. Data Cleaning

20. What is data cleaning?

Answer:
Data cleaning means identifying and correcting bad, incomplete, duplicate or inconsistent data.

---

21. What are some common data-cleaning operations?

Answer:

Examples include:

- Remove duplicate rows
- Remove null values
- Correct data types
- Trim spaces
- Replace incorrect values
- Standardize text
- Handle missing values
- Remove unnecessary columns

---

22. Why are correct data types important?

Answer:
Correct data types are important because calculations, relationships, filters and aggregations depend on them.

For example, if a sales amount is stored as text, Power BI cannot correctly calculate its sum.

G. Looker Studio as a GCP BI Layer

23. What is Looker Studio?

Answer:
Looker Studio is Google's cloud-based reporting and visualization tool. It can connect to sources such as BigQuery, Google Sheets and databases and create interactive reports and dashboards.

---

24. Why is Looker Studio commonly used with BigQuery?

Answer:
BigQuery stores and processes large analytical datasets, while Looker Studio provides the visualization layer.

The architecture can be:

Source Systems

↓

BigQuery

↓

Looker Studio

↓

Dashboard

↓

Business Users

---

25. Is Looker Studio a data warehouse?

Answer:
No. Looker Studio is mainly a reporting and visualization tool.

BigQuery can act as the analytical data warehouse.

---

26. What is a Looker Studio data source?

Answer:
A data source represents the connection between a Looker Studio report and the underlying system containing the data.

For example:

Looker Studio

↓

BigQuery Connector

↓

sales_dataset.sales

H. BigQuery Reporting

27. What is BigQuery reporting?

Answer:
BigQuery reporting means using data stored or processed in BigQuery as the backend for analytical reports and dashboards.

BI tools such as Looker Studio, Power BI and Tableau can query BigQuery.

---

28. Can Power BI connect to BigQuery?

Answer:
Yes. Power BI has a Google BigQuery connector that can be used to access BigQuery datasets and tables.

---

29. How can BigQuery improve reporting performance?

Answer:
We can improve reporting performance using:

- Partitioned tables
- Clustered tables
- Pre-aggregated tables
- Materialized views
- Query filtering
- Avoiding SELECT *
- Proper SQL design

I. Looker vs Looker Studio

30. What is the difference between Looker and Looker Studio?

Answer:

Looker is an enterprise BI and semantic modeling platform.

Looker Studio is mainly a reporting and dashboard creation platform.

Looker provides features such as:

- LookML
- Semantic modeling
- Centralized business definitions
- Governance

Looker Studio focuses more on:

- Visualization
- Charts
- Reports
- Dashboards

---

31. What is LookML?

Answer:
LookML is Looker's modeling language used to define dimensions, measures, joins and business logic.

---

32. Does Looker Studio use LookML?

Answer:
No. LookML belongs to **Looker**, not Looker Studio.

J. Designing Schemas

33. What is a database schema?

Answer:
A schema represents the logical structure of data, including tables, columns, relationships and constraints.

---

34. What is a star schema?

Answer:
A star schema is a dimensional modeling structure where a central fact table is connected to dimension tables.

Example:

```
          DIM_DATE
            |
DIM_CUSTOMER -- FACT_SALES -- DIM_PRODUCT
            |
          DIM_STORE
```

---

35. What is a fact table?

Answer:
A fact table stores measurable business transactions.

Examples:

- Sales amount
- Quantity
- Discount
- Profit

---

36. What is a dimension table?

Answer:
A dimension table stores descriptive information used to analyze facts.

Examples:

- Customer
- Product
- Date
- Region
- Store

---

37. Why is star schema recommended in Power BI?

Answer:
Star schema simplifies relationships, improves performance and makes DAX calculations easier to understand.

---

38. What is the difference between star and snowflake schemas?

Answer:

A **star schema** contains relatively denormalized dimension tables.

A **snowflake schema** further normalizes dimensions into additional related tables.

K. Writing Queries

39. What is SQL?

Answer:
SQL stands for Structured Query Language. It is used to retrieve, insert, update, delete and manipulate relational database data.

---

40. What is a SELECT statement?

Answer:
SELECT retrieves data from a table.

Example:

SELECT customer_name, sales

FROM sales;

---

41. What is WHERE?

Answer:
WHERE filters individual rows before aggregation.

SELECT *

FROM sales

WHERE region = 'West';

---

42. What is GROUP BY?

Answer:
GROUP BY groups rows so aggregate functions can be calculated.

SELECT region, SUM(sales)

FROM sales

GROUP BY region;

---

43. What is HAVING?

Answer:
HAVING filters results after aggregation.

SELECT region, SUM(sales)

FROM sales

GROUP BY region

HAVING SUM(sales) > 100000;

L. DAX – Data Analysis Expressions

44. What is DAX?

Answer:
DAX stands for Data Analysis Expressions. It is the formula language used by Power BI for creating measures, calculated columns and calculated tables.

---

45. Where is DAX used?

Answer:

DAX is mainly used in:

- Power BI
- Power Pivot

- Analysis Services Tabular models

---

46. What is a measure?

Answer:
A measure is a dynamic calculation evaluated based on the current filter context.

Example:

Total Sales =

SUM(Sales[SalesAmount])

---

47. What is a calculated column?

Answer:
A calculated column calculates a value for each row and stores the result in the model.

Example:

Total Amount =

Sales[Quantity] * Sales[Price]

---

48. Measure vs calculated column?

Answer:

A calculated column is evaluated **row by row and stored**.

A measure is calculated **when required according to filter context**.

M. DAX Aggregation

49. What are DAX aggregation functions?

Answer:

Common aggregation functions are:

SUM()

AVERAGE()

MIN()

MAX()

COUNT()

COUNTROWS()

DISTINCTCOUNT()

---

50. Write DAX for total sales.

Total Sales =

SUM(Sales[SalesAmount])

---

51. Write DAX for average sales.

Average Sales =

AVERAGE(Sales[SalesAmount])

---

52. How do you calculate distinct customers?

Total Customers =

DISTINCTCOUNT(Sales[CustomerID])

N. DAX Statistics

53. What statistical functions are available in DAX?

Answer:

Some common functions include:

AVERAGE()

MEDIAN()

MIN()

MAX()

STDEV.P()

STDEV.S()

VAR.P()

VAR.S()

---

54. How would you calculate average order value?

Average Order Value =

DIVIDE(

[Total Sales],

DISTINCTCOUNT(Sales[OrderID])

)

O. CALCULATE

55. What is CALCULATE in DAX?

Answer:
CALCULATE evaluates an expression after changing the current filter context.

Example:

West Sales =

CALCULATE(

[Total Sales],

Sales[Region] = "West"

)

---

56. Why is CALCULATE important?

Answer:
CALCULATE is one of the most important DAX functions because it allows us to modify filter context.

---

57. What is filter context?

Answer:
Filter context represents the filters applied while a DAX expression is being evaluated.

Filters may come from:

- Slicers
- Visuals
- Page filters
- Report filters
- CALCULATE

---

58. What is row context?

Answer:
Row context means a calculation is being evaluated for a particular row.

Calculated columns typically operate using row context.

P. DAX Studio

59. What is DAX Studio?

Answer:
DAX Studio is an external tool used to write, execute and analyze DAX queries against Power BI, Analysis Services and Power Pivot models.

---

60. Why would you use DAX Studio?

Answer:

DAX Studio is useful for:

- Testing DAX queries
- Performance analysis
- Query timing
- Server timings
- Query plans
- Inspecting model metadata
- Identifying expensive DAX calculations

---

61. What are Server Timings?

Answer:
Server Timings show how long the Power BI storage and formula engines spend processing a query.

They help identify performance bottlenecks.

---

62. What are the Formula Engine and Storage Engine?

Answer:

The **Storage Engine** retrieves and aggregates compressed data.

The **Formula Engine** evaluates more complex DAX logic.

Efficient DAX attempts to allow as much work as possible to be handled by the Storage Engine.

Q. Basic DAX Query Writing

63. What is EVALUATE in DAX Studio?

Answer:
EVALUATE is used to return a table from a DAX query.

Example:

```
EVALUATE

Sales
```

---

64. How do you summarize data in DAX?

```
EVALUATE

SUMMARIZECOLUMNS(

    Product[Category],

    "Sales", [Total Sales]

)
```

R. Performance Monitoring Fundamentals

65. What is report performance monitoring?

Answer:
Report performance monitoring means measuring how quickly data loads, DAX calculations execute and visuals render.

---

66. What can make Power BI reports slow?

Answer:

Common causes include:

- Poor data model
- Too many columns
- High-cardinality fields
- Complex DAX
- Too many visuals
- Bidirectional relationships
- Poor DirectQuery SQL
- Large unoptimized source tables

---

67. What is Performance Analyzer?

Answer:
Performance Analyzer is a Power BI Desktop feature that measures how long each visual takes to execute its DAX query and render.

S. Report Generation

68. What is a Power BI report?

Answer:
A Power BI report is a multi-page interactive collection of visualizations built from a semantic model.

---

69. Report vs dashboard?

Answer:

A **report** can contain multiple pages and supports detailed interaction.

A **dashboard** is normally a single-page canvas created in Power BI Service from pinned report visuals.

T. Creating Reports

70. What elements can a Power BI report contain?

Answer:

Examples include:

- Bar charts
- Column charts
- Line charts
- Cards
- KPI visuals
- Tables
- Matrices
- Maps
- Slicers
- Scatter plots

U. Custom Visuals

71. What is a custom visual?

Answer:
A custom visual is a visualization that extends Power BI beyond the standard built-in visuals.

Custom visuals can be obtained from Microsoft AppSource or developed using the Power BI visual SDK.

---

72. Why use custom visuals?

Answer:
They may be used when the required visualization is not available in the standard visual set.

However, security, performance and organizational approval should be considered.

V. Slicing and Filtering

73. What is a slicer?

Answer:
A slicer is a visual control that allows report users to interactively filter data.

Examples:

Region

Product

Year

Customer

---

74. What types of filters are available in Power BI?

Answer:

- Visual-level filter
- Page-level filter
- Report-level filter
- Drillthrough filter
- Slicer

---

75. What is the difference between a slicer and a filter?

Answer:
Both filter data, but a slicer is visible on the report canvas and intended for user interaction.

The filter pane may be visible or hidden depending on report configuration.

W. Conditional Formatting

76. What is conditional formatting?

Answer:
Conditional formatting dynamically changes formatting based on values or rules.

For example:

Profit > 0 → positive indicator

Profit < 0 → negative indicator

---

77. Where can conditional formatting be used?

Answer:

It can be applied to:

- Tables
- Matrices
- Data bars
- Background formatting
- Font formatting
- Icons

X. Analyze Feature

78. What is the Analyze feature in Power BI?

Answer:
Power BI provides analytical capabilities that help investigate why a value increased or decreased and identify factors influencing a metric.

---

79. What does "Explain the increase" do?

Answer:
It analyzes underlying data to identify fields that may explain why a measure increased between selected points.

Y. Detecting Outliers

80. What is an outlier?

Answer:
An outlier is a data point that is significantly different from the majority of observations.

Example:

Typical order value:

$100–$500

One order:

$50,000

The $50,000 order may be an outlier.

---

81. How can you detect outliers in BI reporting?

Answer:

We can use:

- Scatter plots
- Box plots
- Standard deviation
- Z-score
- Percentiles
- Power BI anomaly detection
- Statistical analysis

---

82. Why should outliers be investigated?

Answer:
An outlier may represent:

- A data-quality problem
- Fraud
- Exceptional customer behavior
- System error
- Genuine business opportunity

Therefore it should not automatically be deleted.

Z. Dashboard

83. What is a Power BI dashboard?

Answer:
A dashboard is a single-page visualization canvas in Power BI Service created by pinning visuals from one or more reports.

---

84. Can dashboards be created in Power BI Desktop?

Answer:
No. Traditional Power BI dashboards are created in Power BI Service.

Power BI Desktop creates reports.

AA. Data Alerts

85. What is a Power BI data alert?

Answer:
A data alert notifies a user when a dashboard value crosses a configured threshold.

Example:

Alert me when:

```
Sales < $100,000
```

---

86. Where are Power BI data alerts configured?

Answer:
They are configured in Power BI Service on supported dashboard tiles.

AB. Dataset / Semantic Model Refresh

87. What is dataset refresh?

Answer:
Dataset refresh reloads data from the underlying data sources into the Power BI semantic model.

---

88. Why is refresh required in Import mode?

Answer:
Because Import mode stores a copy of source data inside Power BI. If the source changes, Power BI does not automatically know about the change until the model is refreshed.

---

89. What is scheduled refresh?

Answer:
Scheduled refresh allows Power BI Service to automatically refresh an imported semantic model according to a configured schedule.

---

90. What is an on-premises data gateway?

Answer:
An on-premises data gateway provides a secure bridge between cloud services such as Power BI Service and data sources located inside a private or on-premises network.

AC. Streamlit Introduction

91. What is Streamlit?

Answer:
Streamlit is an open-source Python framework used to create interactive web applications and data applications using Python.

---

92. Why is Streamlit popular with data teams?

Answer:

Because developers can build interactive applications using mainly Python without requiring extensive HTML, CSS or JavaScript.

---

93. Explain Streamlit architecture.

A simplified architecture is:

Browser

↓

Streamlit Web App

↓

Python Application

↓

Database / API / Data Warehouse

For example:

User

↓

Streamlit

↓

BigQuery

↓

Analytics Data

---

94. What happens when a Streamlit widget changes?

Answer:
By default, Streamlit reruns the Python script from top to bottom when users interact with widgets.

Caching and session state can help manage the impact of reruns.

AD. Streamlit Virtual Environment

95. Why should Streamlit be installed inside a virtual environment?

Answer:
A virtual environment isolates project dependencies from other Python projects and prevents package-version conflicts.

---

96. How do you create a Python virtual environment?

python -m venv .venv

Activate on Windows:

.venv\Scripts\activate

Install Streamlit:

pip install streamlit

Run:

streamlit run app.py

AE. Streamlit Core Components

97. What does st.dataframe() do?

Answer:
It displays data such as a Pandas DataFrame as an interactive table.

st.dataframe(df)

---

98. What is st.metric()?

Answer:
st.metric() displays KPI-style information.

```
st.metric(
    "Revenue",
    "$250,000"
)
```

---

99. What is st.sidebar()?

Answer:
It allows controls to be placed inside the application's sidebar.

Example:

```
category = st.sidebar.selectbox(
    "Category",
    categories
)
```

---

100. What is st.chart()?

Answer:
Current Streamlit applications generally use chart functions such as:

st.line_chart()

st.bar_chart()

st.area_chart()

or integrations such as Altair and Plotly.

For example:

```
st.bar_chart(df)
```

AF. Connecting Streamlit to Snowflake

101. Can Streamlit connect to Snowflake?

Answer:
Yes. Streamlit can connect to Snowflake using the Snowflake Python connector, Snowpark or configured Streamlit connections.

---

102. What is Snowpark?

Answer:
Snowpark is Snowflake's developer framework that allows processing Snowflake data using languages such as Python while executing much of the computation inside Snowflake.

---

103. How should Snowflake credentials be handled in Streamlit?

Answer:
Credentials should not be hard-coded in source code.

They should be stored in secure configuration such as:

secrets.toml

or environment variables / secret management services.

AG. BigQuery Python Client

104. What is the BigQuery Python client?

Answer:
The BigQuery Python client library allows Python applications to communicate with Google BigQuery.

Installation:

```
pip install google-cloud-bigquery
```

---

105. How do you create a BigQuery client?

```
from google.cloud import bigquery


client = bigquery.Client()
```

---

106. How do you execute a BigQuery query using Python?

```
query = """
SELECT category, SUM(amount)
FROM `project.dataset.sales`
GROUP BY category
"""


result = client.query(query).to_dataframe()
```

---

107. What is Application Default Credentials?

Answer:
Application Default Credentials, or ADC, is a Google Cloud authentication mechanism that allows client libraries to automatically locate appropriate credentials from the environment.

AH. Caching BigQuery in Streamlit

108. Why is caching important in Streamlit?

Answer:
Because Streamlit reruns scripts during user interaction. Without caching, the same expensive BigQuery query may run repeatedly.

Caching improves:

- Performance
- Response time
- Database load
- BigQuery cost

---

109. What is st.cache_data?

Answer:
st.cache_data caches the returned data from functions.

Example:

```
@st.cache_data(ttl=300)

def load_data():

    return client.query(sql).to_dataframe()
```

---

110. What is TTL?

Answer:
TTL means **Time To Live**.

For example:

ttl=300

means the cached result is valid for approximately five minutes.

---

111. What is st.cache_resource?

Answer:
st.cache_resource is used to cache reusable resources such as:

- Database connections
- API clients
- Machine-learning models

Example:

```
@st.cache_resource

def get_client():

    return bigquery.Client()
```

---

112. Difference between st.cache_data and st.cache_resource?

Answer:

st.cache_data is used mainly for **data results**.

```
@st.cache_data

def load_sales():

    ...
```

st.cache_resource is used for **shared resources**.

```
@st.cache_resource

def get_client():

    ...
```

AI. Service Authentication in Streamlit Cloud

113. What is service account authentication?

Answer:
A service account is a non-human Google Cloud identity used by applications to access GCP resources.

For example:

Streamlit Application

↓

Service Account

↓

IAM Permission

↓

BigQuery

---

114. Why use a service account for a deployed Streamlit application?

Answer:
Because the deployed application cannot normally depend on a developer's local Google login. A service account provides application-level authentication.

---

115. What permissions might a Streamlit BigQuery application require?

Answer:
Depending on its purpose, it could require permissions such as:

- BigQuery Job User
- BigQuery Data Viewer

Permissions should follow the principle of least privilege.

---

116. What is the principle of least privilege?

Answer:
Least privilege means giving an identity only the permissions required to complete its tasks and no unnecessary additional privileges.

PART 2 — SCENARIO-BASED INTERVIEW QUESTIONS & ANSWERS

Power BI Scenarios

117. You have a 50-MB sales CSV that changes once every day. Import or DirectQuery?

Answer:
I would normally use **Import mode** because the dataset is relatively small and changes only daily. Import mode should provide faster report performance, and I can configure a daily refresh.

---

118. Your source contains hundreds of millions of rows and business users require near-current information. What would you consider?

Answer:
I would evaluate DirectQuery or a composite architecture rather than simply importing everything.

I would also optimize:

- Source tables
- Partitioning

- Aggregations
- Required columns
- SQL queries

The final decision depends on latency, performance and feature requirements.

---

119. A Power BI report takes 15 seconds to open. What would you investigate?

Answer:
I would check:

1. Number of visuals
2. Data model design
3. Relationship configuration
4. DAX complexity
5. High-cardinality columns
6. Source query performance
7. Import versus DirectQuery
8. Performance Analyzer
9. DAX Studio Server Timings

---

120. A sales column is being imported as text. What problem can this create?

Answer:
Aggregations such as SUM may not work correctly.

I would correct the data type in Power Query before loading the model.

---

121. You receive customer data with duplicate records. Where would you clean it?

Answer:
If possible, I would correct the issue upstream. For reporting preparation, I could remove or handle duplicates in Power Query after understanding what defines a true duplicate.

Schema Scenarios

122. You have Sales, Customer, Product and Date data. How would you model it?

Answer:

I would preferably create:

```
          DIM_DATE
             |
DIM_CUSTOMER -- FACT_SALES -- DIM_PRODUCT
```

The fact table contains:

Quantity

Revenue

Discount

Profit

The dimension tables contain descriptive information.

---

123. Why would you avoid putting everything into one huge Power BI table?

Answer:
A single flat table can introduce duplicate data, poor maintainability and inefficient models.

A star schema provides clearer relationships and generally improves usability and performance.

DAX Scenarios

124. The manager asks for total revenue. What DAX would you write?

```
Total Revenue =
SUM(Sales[Revenue])
```

---

125. The manager asks for revenue only in the West region.

```
West Revenue =
CALCULATE(
   [Total Revenue],
   Customer[Region] = "West"
)
```

---

126. The manager asks for the number of unique customers.

```
Unique Customers =
DISTINCTCOUNT(Sales[CustomerID])
```

---

127. The manager wants average sales per order.

```
Average Order Value =

DIVIDE(

   [Total Revenue],

   DISTINCTCOUNT(Sales[OrderID])

)
```

---

128. Why would you choose a measure instead of a calculated column for total sales?

Answer:
Total sales is an aggregation that should respond dynamically to filters such as year, region and product.

Therefore, a measure is more appropriate.

---

129. A calculated column returns the same value repeatedly. What would you check?

Answer:
I would check whether the calculation should actually be a measure and whether the DAX expression is correctly using row context.

---

130. Your manager asks for total sales regardless of product selection. How would you approach it?

Answer:

I can modify the filter context using CALCULATE and functions such as REMOVEFILTERS.

```
All Product Sales =

CALCULATE(

   [Total Sales],

   REMOVEFILTERS(Product)

)
```

DAX Performance Scenarios

131. One DAX measure is extremely slow. How would you troubleshoot it?

Answer:
I would use:

- Performance Analyzer
- DAX Studio

- Server Timings
- Query Plan

Then I would investigate:

- Expensive iterators
- Complex FILTER operations
- Large intermediate tables
- Poor relationships
- High-cardinality columns

---

132. What if SUMX is slow on a very large table?

Answer:
I would investigate whether the calculation can be simplified into an aggregation or moved upstream.

For example, instead of repeatedly iterating:

```
SUMX(
   Sales,
   Sales[Quantity] * Sales[Price]
)
```

it may be appropriate in some models to calculate or precompute the value earlier.

The correct approach depends on the model and business logic.

Reporting Scenarios

133. Your executive wants only five major business metrics. Would you create a complex detailed report?

Answer:
No. I would create a simple executive view containing important KPIs such as:

Revenue

Profit

Customers

Orders

Growth

Detailed analysis could be provided on additional report pages.

---

134. When would you use a table rather than a chart?

Answer:
I would use a table when users need exact detailed values.

A chart is better when users need to quickly identify trends, comparisons or patterns.

---

135. When would you use a line chart?

Answer:
A line chart is appropriate for showing trends over time.

For example:

Monthly Revenue

Daily Website Visits

Quarterly Profit

---

136. When would you use a bar chart?

Answer:
A bar chart is useful for comparing values across categories.

For example:

Revenue by Product

Sales by Region

Profit by Department

Filtering Scenarios

137. The user wants to select one region and have every visual change. What would you use?

Answer:
I would use a Region slicer and configure visual interactions appropriately.

---

138. One visual should not respond to a slicer. What would you do?

Answer:
I would use **Edit interactions** and disable the slicer's filtering effect on that visual.

---

139. A filter should affect only one page. Which filter would you use?

Answer:
A page-level filter.

---

140. A filter should affect every page in the report.

Answer:
I would use a report-level filter or a synchronized slicer depending on the user-experience requirement.

Conditional Formatting Scenario

141. Management wants negative profit values highlighted. How would you implement it?

Answer:
I would configure conditional formatting based on Profit.

For example:

Profit < 0 → warning formatting

Profit > 0 → positive formatting

Outlier Scenarios

142. One transaction is ten times larger than normal. Would you delete it?

Answer:
No. I would first investigate it.

It could be:

- Genuine large transaction
- Duplicate record
- Data-entry error
- Fraud
- Incorrect unit conversion

Outliers should be investigated before removal.

---

143. How could you visually identify outliers?

Answer:
I could use a scatter plot or statistical measures such as standard deviation, percentiles or Z-score.

Dashboard Scenarios

144. A manager wants one page showing information from multiple reports. What Power BI feature could help?

Answer:
A Power BI Service dashboard can pin tiles from different reports and provide a consolidated overview.

Alert Scenarios

145. Management wants notification when daily sales drop below $50,000.

Answer:
If the required metric is supported as an alertable dashboard tile, I could configure a Power BI Service data alert.

In larger workflows, Power Automate or monitoring systems may also be appropriate.

Refresh Scenarios

146. Your source database updates every night at 1 AM. When should Power BI refresh?

Answer:
I would schedule the refresh after the source load completes successfully, for example around 2 AM depending on the expected load duration.

This avoids refreshing while data is incomplete.

---

147. A scheduled refresh suddenly fails. What would you check?

Answer:

I would investigate:

- Credentials
- Gateway availability
- Source connectivity
- Schema changes
- Query errors
- Refresh history
- Source system availability
- Timeout or capacity problems

---

148. The on-premises database works from Power BI Desktop but refresh fails in Power BI Service.

Answer:
I would check whether the on-premises data gateway is installed, online and correctly mapped to the data source.

Looker Studio / BigQuery Scenarios

149. Your company stores analytics data in BigQuery and wants a lightweight dashboard. What could you use?

Answer:

A straightforward architecture would be:

BigQuery

↓

Looker Studio

↓

Interactive Dashboard

Looker Studio can connect directly to BigQuery.

---

150. A Looker Studio dashboard connected to BigQuery is expensive to use. What would you investigate?

Answer:

I would investigate:

- Amount of data scanned
- Query frequency
- Dashboard filters
- Repeated queries
- Partitioning
- Clustering
- Pre-aggregation
- Materialized views
- BI Engine where appropriate

---

151. Should you use Looker or Looker Studio when you require centralized business definitions and governed metrics?

Answer:
Looker would generally be the stronger choice because it provides a semantic modeling layer using LookML and stronger enterprise governance capabilities.

Streamlit Scenarios

152. You need to create a Python dashboard quickly without writing JavaScript. What would you choose?

Answer:
Streamlit would be a strong option because the UI can be created directly from Python.

---

153. You want to display a Pandas DataFrame.

```
st.dataframe(df)
```

---

154. You want to show total revenue as a KPI.

```
st.metric(
    "Total Revenue",
    "$1.2M"
)
```

---

155. You want filters to appear on the left side.

```
region = st.sidebar.selectbox(
    "Region",
    regions
)
```

Streamlit Rerun Scenario

156. Every time a dropdown changes, your Streamlit SQL query executes again. Why?

Answer:
Streamlit normally reruns the script when user interaction changes widget state.

Therefore database-query functions should be designed carefully and caching may be appropriate.

BigQuery + Streamlit Scenarios

157. Your Streamlit app connects to BigQuery. What architecture would you explain?

User Browser

↓

Streamlit Application

↓

BigQuery Python Client

↓

BigQuery

↓

Query Result

↓

Pandas DataFrame

↓

Streamlit Chart

---

158. The user changes chart type but not the underlying data. Should BigQuery execute again?

Answer:
Ideally, not necessarily.

If the underlying query parameters have not changed, I would cache the query result and reuse the data.

Caching Scenarios

159. You have this function:

```
@st.cache_data(ttl=300)

def load_data():

    ...
```

What does it mean?

Answer:
The returned data is cached, and repeated calls with the same relevant function inputs can reuse the cached result until the cache expires or is invalidated.

300 seconds equals five minutes.

---

160. The user selects "Electronics" twice within five minutes. Should BigQuery execute twice?

Suppose:

@st.cache_data(ttl=300)

def load_data(category):

   ...

and both calls are:

load_data("Electronics")

Answer:
Normally, the second call can use the cached result because the function and arguments are the same and the TTL has not expired.

---

161. The user selects Electronics and then Clothing. Will the same cache be used?

Answer:
No. Different function arguments create different cache entries.

Conceptually:

load_data("Electronics")

    ↓

Cache A

load_data("Clothing")

    ↓

Cache B

---

162. Why cache the BigQuery client with st.cache_resource?

Answer:
Creating a reusable client repeatedly is unnecessary.

I can use:

@st.cache_resource

def get_client():

   return bigquery.Client()

The same resource can then be reused across reruns.

---

163. Should I use st.cache_resource for a DataFrame query result?

Answer:
Normally no.

I would use:

st.cache_data

for query results and:

st.cache_resource

for shared resources such as database clients.

Authentication Scenarios

164. Your Streamlit app works locally with your Google login but fails after deployment. Why?

Answer:
Locally, the application may be using my personal Application Default Credentials.

The deployed environment does not automatically have those credentials.

I would configure appropriate service authentication for the deployed application.

---

165. Would you store this inside GitHub?

```
SERVICE_ACCOUNT_KEY = {

   "private_key": "..."

}
```

Answer:
No.

Credentials should not be committed to source-control repositories.

They should be managed using a secrets mechanism or secure cloud identity mechanism.

---

166. Your Streamlit application only needs to read BigQuery tables. Should it receive BigQuery Admin?

Answer:
No.

That violates least privilege.

I would give only the permissions required for query execution and data access.

---

167. Why might both BigQuery Job User and BigQuery Data Viewer be required?

Answer:

They solve different requirements.

BigQuery Job User allows the identity to create query jobs.

BigQuery Data Viewer allows the identity to read table data.

A query application may require both depending on where permissions are assigned.

MULTI-TOPIC REAL-WORLD INTERVIEW SCENARIOS

168. Design a complete BI solution for a retail company using BigQuery and Power BI.

Answer:

I could propose:

POS / E-commerce / CRM

↓

Data Pipeline

↓

BigQuery

↓

Reporting Views

↓

Power BI

↓

Business Reports

I would:

1. Load source data into BigQuery.
2. Transform raw data into reporting tables.
3. Design fact and dimension structures.
4. Connect Power BI to BigQuery.
5. Build measures using DAX.
6. Create reports and slicers.
7. Publish to Power BI Service.
8. Configure refresh or DirectQuery according to the requirement.
9. Configure security.
10. Monitor performance.

169. **Design the same solution using Streamlit.**

Answer:

BigQuery

↓

BigQuery Python Client

↓

Streamlit

↓

Cached Query Results

↓

Filters + Metrics + Charts

↓

Users

Example structure:

```
@st.cache_resource
def get_client():
    return bigquery.Client()



@st.cache_data(ttl=300)
def load_sales(category):
    client = get_client()


    sql = """
    SELECT category, SUM(amount) revenue
    FROM `project.dataset.sales`
    WHERE category = @category
    GROUP BY category
    """
```

    return ...

The deployed application would authenticate using an appropriate application identity with least-privilege IAM permissions.

---

170. Business asks: "Power BI, Looker Studio or Streamlit — which should we use?"

Answer:
It depends on the requirement.

I would consider **Power BI** when the organization uses the Microsoft ecosystem and needs strong self-service BI, modeling and DAX capabilities.

I would consider **Looker Studio** when a lightweight dashboarding solution integrates naturally with Google services such as BigQuery.

I would consider **Streamlit** when we need a custom Python-driven analytical application rather than only a traditional BI report.

---

171. Your BigQuery dashboard is simultaneously slow and expensive. Give an end-to-end optimization strategy.

Answer:

I would investigate from both the database and visualization sides.

BigQuery

↓

Partitioning

↓

Clustering

↓

Optimized SQL

↓

Pre-aggregation

↓

Only required columns

↓

Caching

↓

Optimized BI visuals

I would avoid:

SELECT *

when unnecessary, reduce data scans, use appropriate filters and prevent the frontend from unnecessarily executing identical queries repeatedly.

---

172. The source has 100 million rows but the dashboard needs only monthly sales by region. Would you transfer all 100 million rows?

Answer:
No.

I would preferably aggregate the data close to the source.

For example:

```
SELECT
   DATE_TRUNC(order_date, MONTH) AS month,
   region,
   SUM(sales_amount) AS revenue
FROM sales
GROUP BY
   month,
   region;
```

The reporting layer then receives only the data required for analysis.

---

173. A manager says, "The number on my dashboard is wrong." What would you investigate?

Answer:

I would trace the metric from end to end:

Source Data

↓

Transformation

↓

Data Model

↓

Relationships

↓

DAX / SQL

↓

Filters

↓

Visual

I would verify the business definition first.

For example, "Total Customers" could mean:

COUNT(CustomerID)

or

DISTINCTCOUNT(CustomerID)

Those calculations may produce very different answers.

---

174. What would you say if the interviewer asks, "How do you build a good dashboard?"

Answer:
I first understand the business question and identify the KPIs. Then I prepare a clean data model, create validated calculations, select visuals that match the analytical requirement, provide useful filters, keep the dashboard simple, test performance and validate the final numbers against the source data.

A strong flow is:

Business Requirement

↓

Understand Data

↓

Clean Data

↓

Design Model

↓

Create Measures

↓

Choose Visuals

↓

Add Filters

↓

Validate Numbers

↓

Optimize

↓

Publish
